Sparse syntax and weighted Pell transport

Explicit Diophantine equations for the consistency of ZFC

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Abstract

We give two explicit improvements to the inspected ProveIt constructions. The direct construction uses a sparse, non-surjective coding of its numerical proof calculus. A finite-marker substitution theorem proves that the unused numeric codes do not change finite refutability. Combining this coding with a signed affine horizon and divisibility-forced Pell growth produces a quartic in 208 unrestricted integer variables, specified by 148 quadratic residuals. The separate compact construction gives a polynomial of exact degree 190 in ten strictly positive integer variables, with 120 binary arithmetic operations and a completely specified fixed coefficient tuple. Its new weight b^2UE permits a whole-product Pell center in a smaller arithmetic chart, including an auxiliary index previously excluded in the other chart. For each construction, ordinary proofs of the new steps compose with explicitly identified inherited calibrations to prove in ZFC that absence of solutions is equivalent to syntactic Con(ZFC). All coefficients, evaluators, exact finite checks, source bindings, and editable article sources accompany the paper. Neither global minimality nor a new end-to-end proof-assistant verification is claimed.

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1 Results, comparison, and the precise logical claim

Fix the effective ZFC presentation used in the repository's retained calibration: its axiom predicate is `ZFCax_s`, its finite derivation relation is `BProv`, and its false formula is `Form.fBot`. Write $\mathrm{Ref}_{\mathrm{ZFC}}$ for the arithmetic assertion that a finite refutation in this presentation exists, and put $\mathrm{Con}(\mathrm{ZFC}) = \neg\,\mathrm{Ref}_{\mathrm{ZFC}}$.

Our comparison snapshot is the public ProveIt commit `d123610bc98fef2006bff1c9466bc3f1221003e7`. We inspected the supplied Hilbert Tenth Problem directory, the numbered consistency constructions through package 178, and the seven archives then in `docs/incoming`. The incoming Pell Growth 210 archive supplies the immediate direct parent[2]. Packages 172 and 178, together with the incoming affine-horizon archive, supply the compact comparison and its two principal ingredients[4, 5]. These are comparisons against identified files; they are not a proof of priority or an exhaustive worldwide search.

Construction and domain	Variables	Degree	Rows or operations	Height
Incoming Pell Growth, $\mathbb{Z}$	210	4	150 rows	2^{40}
Affine-horizon companion here, $\mathbb{Z}$	209	4	149 rows	2^{44}
Sparse syntax here, $\mathbb{Z}$	**208**	**4**	**148 rows**	2^{44}
Incoming free chart, $\mathbb{Z}_{>0}$	10	206	90 operations	fixed tuple
Package 178 whole center, $\mathbb{Z}_{>0}$	10	208	116 operations	fixed tuple
Weighted transport here, $\mathbb{Z}_{>0}$	**10**	**190**	**120 operations**	fixed tuple

Height for the direct constructions means the largest absolute coefficient after the single quartic has been expanded and like monomials collected. The new direct quartic has 5,987 collected monomials; its parent has 5,282. The companion has 5,486. The displayed tradeoffs are intentional: dimension, degree, expanded height, support, and arithmetic-circuit length are different measures of simplicity.

Theorem 1.1 (Delivered equations)**.** *Let $R_0, \dots, R_{147}$ be the literal quadratic residuals in Appendix A, and define*

$$Q_{208}(x_0, \dots, x_{207}) = \sum_{j=0}^{147} R_j(x_0, \dots, x_{207})^2.$$

Let P be the ten-variable polynomial in Section 5, using the four fixed integers completely defined in Appendix B. Then ordinary mathematics formalizable in ZFC proves

$$\begin{aligned} \exists \boldsymbol{x} \in \mathbb{Z}^{208}\ Q_{208}(\boldsymbol{x}) = 0 \quad &\longleftrightarrow \quad \mathrm{Ref}_{\mathrm{ZFC}}, \\ \exists \boldsymbol{u} \in \mathbb{Z}_{>0}^{10}\ P(\boldsymbol{u}) = 0 \quad &\longleftrightarrow \quad \mathrm{Ref}_{\mathrm{ZFC}}\,. \end{aligned}$$

Their negations are therefore ZFC-provably equivalent to $\mathrm{Con}(\mathrm{ZFC})$*. The first equation has exact degree four and the second has exact degree 190.*

The quantifiers in the first assertion range independently over all integers. The second assertion has a strictly positive domain. Its named registers are abbreviations for polynomials in ten inputs, so they are not extra existential witnesses. The fixed coefficient recipe uses finite sums, digit extraction, and fixed powers only to name four particular integers. No variable exponentiation or floor occurs in either delivered polynomial.

The direct route evaluates the numerical calculus C222 itself and uses its identified C222/ZF/ZFC calibration. It does not simulate K215 or specialize Jones's universal polynomial. The compact route retains its existing native/K215 ancestry. Allowing alternative sources is useful, but replacing an already effective source is not automatically a simplification.

Proof status. The new results are ordinary mathematical arguments, supported by exact coefficient checks. We have not emitted an object-language ZFC derivation or rebuilt a complete transitive Lean/Rocq dependency closure. The retained logical endpoints are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation` on the direct side, and the explicitly named whole-tuple native/refutation calibration on the compact side. Finite checks authenticate formulas and transformations; they do not prove unbounded Pell or logical theorems by sampling. The result assumes neither consistency nor inconsistency and proves neither one.

2 Sparse syntax: a finite proof does not need a surjective coding

The earlier square-shell coding makes every natural number a constructor value. That property simplifies global recoding, but it is stronger than what is needed to preserve the existence of a finite refutation. Define

$$\begin{aligned} &\pi(x,y) = (x+y)^2 + x, \\ &I(x,y) = 4\pi(x,y), \qquad M(x,y) = 4\pi(x,y) + 1, \\ &A(x,y) = 4\pi(x,y) + 2, \quad E(x,y) = 4\pi(x,y) + 3. \end{aligned} \tag{1}$$

Here I, M, A, E name the implication, membership, quantifier, and equality constructors of the same numerical rule schemata. They are polynomials, not syntactic predicates. Their arguments may be any natural numbers.

For $x+y=s$, $\pi(x,y)$ runs injectively through $[s^2, s^2+s]$. Different shells have disjoint intervals. The four constructor ranges are disjoint by their residues modulo four. The first unused values are $12, 13, 14, 15$, because the quotient 3 is not in the range of π. Every nonzero constructor value is greater than both arguments. The only zero case is $I(0,0) = 0$, and $M(0,0) = 1$.

2.1 Normalized trees and finite markers

Decode a numeral recursively: zero is a distinguished leaf; a nonzero constructor value is its uniquely determined constructor node; a number outside all constructor ranges is a new atomic leaf. Strict descent of the arguments makes this recursion terminate. Trees are normalized by identifying $I(0,0)$ with zero. Apart from this convention, injectivity and disjointness give unique trees.

The reference Cantor constructors are

$$I_o(x,y) = 2(x+y)(x+y+1) + 4y, \qquad M_o = I_o + 1, \quad A_o = I_o + 2, \quad E_o = I_o + 3.$$

They partition $\mathbb{N}$ and have the same descent and zero convention. Every reference numeral therefore has a normalized tree built from zero alone.

Lemma 2.1 (Finite-marker substitution)**.** *Let $\mathcal{S}$ be a finite, subtree-closed set of normalized sparse-code trees. There is a substitution of reference-code numerals for the extra atomic leaves that is injective on $\mathcal{S}$, fixes 0 and 1, and preserves every constructor occurrence represented in $\mathcal{S}$.*

Proof. Let D be the maximum height in $\mathcal{S}$ and let there be k distinct extra leaves. If $k = 0$, evaluate the same trees in the reference coding. Otherwise put $L = \max(D, k)$ and define

$$W_0 = 0, \qquad W_{r+1} = A_o(0, W_r), \qquad m_j = I_o(W_L, W_j) \quad (0 \le j < k).$$

The markers m_j are distinct and all have height $L + 1 > D$. Substitute the jth marker for the jth extra leaf.

A tree without extra leaves has height at most D. The image of one extra leaf has height $L+1$. The image of a constructor tree containing an extra leaf has height greater than $L+1$, with height determined by the deepest occurrence of an extra leaf. Thus a marker cannot collide with a constant-only tree or a nontrivial tree containing markers. If two remaining nonzero constructor trees have equal images, disjoint constructor ranges identify the root symbol and injectivity identifies both child images. Induction then identifies the original trees. The normalization $I(0,0) = 0$ is imposed on both sides, so it introduces no new collision. Substitution fixes zero and $1 = M(0,0)$ and commutes with the constructors by definition. □

Proposition 2.2 (Finite refutation equivalence)**.** *The sparse C222 calculus derives 1 if and only if the reference C222 calculus derives 1. ZFC proves the equivalence by effective finite proof transformations.*

Proof. For a finite sparse derivation, collect every parameter, premise, conclusion, and intermediate value of every instantiated constructor term. Decode these values and include all subtrees. Lemma 2.1 transfers each instantiated rule to a reference rule. It preserves truth at zero, modus ponens, generalization, and all constructor-term axioms. Its injectivity preserves the guarded bundle's two disequalities $c \neq a$ and $c \neq b$. The concluding numeral remains 1.

Conversely, evaluate the unique zero-generated tree of each reference numeral with the sparse constructors. Uniqueness of sparse decoding makes this a global injection on reference numerals. It commutes with all constructors, fixes 0 and 1, and transports every finite derivation and its guards to the sparse calculus.

The arithmetic bundle guard uses the existential identity $((c-a)(c-b))^2 = (d+1)^2$. After translating a guarded instance, choose $d = |(c-a)(c-b)| - 1 \geq 0$. One must not assume that a numeral recoding commutes with addition or preserves the old d. In other rule branches, d is simply a schema parameter and is translated in the ordinary way. These are alternative local branches, so the choices do not conflict. □

There is generally no global injective homomorphism in the reverse direction fixing zero: the reference constructor closure of zero already exhausts the reference numerals. The dependence on the finite derivation is essential. It is also sufficient, because inconsistency is the existence of one finite refutation. This is the principal conceptual change in the direct construction.

3 Signed horizons and a compatible Pell growth certificate

For an integer $a \geq 2$, use the canonical Pell notation

$$(a + \sqrt{a^2-1})^n = X_n(a) + Y_n(a)\sqrt{a^2-1} \qquad (n \in \mathbb{Z}).$$

In the compact construction's inherited notation, $\psi_a(n) = Y_n(a)$.

The arithmetic source uses a horizon t, a growth value F, a beta spacing y, a beta code x, and an even CRT index representative C. They are ordinary integer witnesses before their signs have been proved. Write $d = 2^{19}$. Replace the old positive square slice by the affine slice

$$t = 2 + 4z, \qquad z \in \mathbb{Z}. \tag{2}$$

For the sparse syntax use the three quadratic growth equations

$$Z_g = tF, \qquad X_g^2 - 64Z_g^2 - 16Z_gF = 1, \qquad F = dt(2u+1). \tag{3}$$

The changed center is $8t+1$. Keeping the old center $4t+1$ without updating the label bound would leave a proof gap; the literal final equation contains the new coefficients 64 and 16.

3.1 Recover the sign before classifying the growth Pell equation

The retained source has a common square frame and a largest-index certificate for C. Its inherited theorem recovers each principal Pell pair at its signed nominal index before the front variables are oriented. For a requested base with absolute value at least two, its signed power lemma then forces the exponent to be nonnegative and the output to be the exact power. The actual frame is retained with every requested base, output, and nominal exponent included.

Equation (2) gives even $t \neq 0$, $|t| \geq 2$. The last row of (3), without using its norm row, already gives

$$F \neq 0, \qquad F \equiv 0 \pmod 2, \qquad |F| \geq d|t| \geq 2^{20}.$$

In particular the existing square frame exceeds its required threshold 258, because it includes F^2.

The affine radix is $D_r = 2(C + F + 1)$. Since C and F are even, $|D_r| \geq 2$. The retained request $q = D_r^x$ gives $x \geq 0$. The beta equations are

$$y = F(2v + 1), \qquad x = F + 1 + (y + 1)r_0, \qquad x = 1 + [1 + (t + 1)y]r_t. \tag{4}$$

They imply $|y| \geq |F|$ and

$$|1 + (t + 1)y| \geq |t + 1|\,|y| - 1 \geq |F| - 1 > 1.$$

Consequently $x = 0$ would express -1 as a multiple of a nonunit, which is impossible. Thus $x \geq 1$, $|q| \geq 2$, and the existing request at base q with exponent t forces $t \geq 0$. The affine slice excludes zero, so $t \geq 2$.

This bootstrap uses the elementary divisibility lower bound before the standalone Pell center has been oriented. It never assumes a growth estimate at a negative center. It is the new composition of the signed horizon and the incoming divisibility-growth construction.

3.2 Label bound and growth

For $0 \leq r, s \leq M$, each sparse constructor satisfies

$$C(r, s) + 1 \leq 16M^2 + 4M + 4 \leq 16(M + 1)^2.$$

Every leaf of a local constructor term is below t or is 0 or 1. Induction on depth gives the bound $16^{2^h-1}t^{2^h}$ for a term's value plus one. The deepest source term has depth 19; the eight ordinary axiom terms have depth at most five. Define

$$L = 16^{d-1}t^d.$$

Every required local formula label is less than L.

After the sign bootstrap, $a = 8t + 1 \geq 17$ and the norm row is $X_g^2 - (a^2 - 1)F^2 = 1$. Write $|F| = Y_n(a)$ with $n > 0$ for its canonical Pell ordinate. The retained elementary identities give

$$Y_n(a) \equiv n \pmod{a - 1}, \qquad v_2(Y_n(a)) = v_2(n) \quad (a \text{ odd}).$$

Since $8t \mid F$, one has $8t \mid n$. The exact valuation of $F = dt(2u + 1)$ then shows that $n/(8t)$ is divisible by 2^{16}, and hence $n \geq dt$. The recurrence growth estimate yields

$$|F| \geq (16t + 1)^{dt-1} \geq (16t)^{dt-1} \geq L^t,$$

because the ratio in the last comparison is $16^{t-1}/t \geq 1$. In particular $|F| > t$.

The retained requests at exponents $y + 1$ and $F + t$, together with (4), now recover

$$F > 0, \quad y \geq F, \quad x \geq 2 + (t + 1)y, \quad C > F + t, \quad D_r > 2C, \quad q = D_r^x > 0.$$

The common bound $F \geq L^t$ is exactly what is needed by the inherited finite-product separation and Horn-flow argument. It also gives $F \geq t^{2t} \geq \operatorname{lcm}(1, \ldots, t)^2$, as required by the generalized factorial-free CRT selector.

3.3 An explicit new bound for the changed local polynomial

The syntax change must also respect the modulus used to recover the local row. Put $R_0 = 2+t+x+y+F$. Every decoded local atom is less than R_0, and $R_0 \geq 16$. If two inputs are below R_0^e, a sparse constructor is below $16R_0^{2e} \leq R_0^{2e+1}$. Thus a depth-19 label is below $R_0^{2^{20}-1}$ and a depth-five label is below R_0^{63}.

The eleven unsquared factors in the local axiom/inference product have the following valid exponent bounds: 1 for the truth factor, 66 for each of eight ordinary axiom factors, 10 for modus ponens, and 7 for generalization. Their squared product is below R_0^{1092}. The squared bundle factor is below $R_0^{2097156}$ and its squared guard defect is below R_0^{10}. The joint factor is therefore below $R_0^{2097157}$. Each of the two beta squares is below R_0^{12}. Including them gives

$$\mathrm{Local} < R_0^{2098250} < R_0^{2^{23}}.$$

For $t \geq 2$, one has $L \geq 2^{5d-4}$ and hence $F \geq L^t \geq 2^{10d-8} > 2^{32}$. The recovered inequalities imply $R_0 < 2x$ and $x > 2^{32}$. The function $x/\log_2(2x)$ is increasing beyond this threshold, and at 2^{32} one has $2^{23} \cdot 33 < 2^{32}$. Hence

$$\mathrm{Local} < (2x)^{2^{23}} < 2^x.$$

This supplies a fresh bound for the changed constructors rather than silently reusing the old coefficients' bound. All remaining low-digit estimates use $D_r > 2C$ and the same authenticated index cap.

3.4 Cofinal completeness

Translate a finite ZFC refutation to a finite sparse-C222 refutation by Proposition 2.2 and the inherited calibration. Choose a sufficiently large positive horizon with

$$t \equiv 2 \pmod{2^{23}}.$$

This progression is cofinal and lies in (2). It has $v_2(t) = 1$ and its binary digit in position 20 is zero. Set

$$n_0 = dt, \quad y_0 = Y_{n_0}(8t+1), \quad a_0 = X_{n_0}(8t+1).$$

The congruence gives $t \mid y_0$, and the exact valuation gives $v_2(y_0) = 20$. Together these facts, including the odd part of t, show that $b_0 = y_0/(dt)$ is an odd integer. Also $a_0 \equiv 1 \pmod 4$.

The incoming proof establishes that $s \mapsto Y_s(a_0)$ permutes residues modulo every power of two when $a_0 \equiv 1 \pmod 4$. Choose $N > 23$ with $2^N > t$ and an arbitrarily large positive odd s such that

$$(t/2)b_0 Y_s(a_0) \equiv 1 \pmod{2^{N-20}}.$$

Composition gives a positive ordinate

$$F = Y_{n_0 s}(8t+1) = y_0 Y_s(a_0) \equiv 2^{20} \pmod{2^N},$$

Complete its witnesses with $X_g = X_{n_0 s}(8t+1)$, $Z_g = tF$, and $u = (F/(dt) - 1)/2$. The latter is integral because the quotient is odd, so all three growth rows hold. The binary supports of F and t are disjoint, so $\binom{F+t}{t}$ is odd. This is the required parity for an even CRT anchor, not an optional heuristic property.

Choose the odd multiplier in $y = F(2v+1)$ to contain every odd prime at most t and to be sufficiently large for the finite pending states. The beta moduli, indexed by $1 \leq i < j \leq t+1$, are pairwise coprime: a common prime of $1+iy$ and $1+jy$ would divide $i-j$, hence be at most t, and would also divide y, a contradiction. The subsequent choice order is the beta code x, the selector and CRT moduli, the ordinary CRT representatives, an arbitrarily large even representative C, the radix D_r, its power q, and finally the shared Pell frame. No later choice determines a parameter on which an earlier modulus depends.

The retained arithmetic completeness theorem then supplies every remaining witness. The local syntax block is regenerated with its new constructors, its labels are covered by L, and its modulus bound was proved above. This establishes the sparse source's existence equivalence with the same finite refutation predicate.

4 From a semantic source to the literal quartic

The source change and the polynomial reductions perform different jobs. The former changes the numeric proof coding and horizon; its justification is the unbounded mathematics of Sections 2–3. The latter are exact finite transformations of the newly defined residuals.

The local compiler represents every required constructor polynomial by quadratic multiplication equations. Its common-expression rule is slightly stronger than syntactic equality. If two quadratic expressions have proportional homogeneous quadratic parts, with integer ratio c, then

$$q_2 = cp_2, \qquad w = p_2 + p_1 \quad \Longrightarrow \quad q_2 + q_1 = cw + (q_1 - cp_1).$$

Here the subscripts 2 and 1 indicate the quadratic and affine parts. The right side is affine in the existing wire. It therefore saves a wire without division or a new integrality condition. The constructor terms and all retained beta, CRT, and Pell rows are separately bound by exact source certificates.

The graft's boundary is especially small: only the joint bundle/guard square and the square of the eleven-factor axis product are passed from the local block to the retained local-row equation. The independent audit checks all nine axiom constructor expressions, the two inference factors, both boundary values, and the retained-row partition. A stale unused definition of FS can be deleted or projected away; it cannot be counted as a necessary witness.

The affine horizon eliminates its own t coordinate. The common-Hessian projections and subsequent integer coordinate/row changes have exact restoration certificates. Coordinate shears and translations preserve integer solutions bijectively. An invertible change of residual basis preserves their simultaneous zero set and therefore preserves the zero set of their sum of squares, although it need not preserve the sum-of-squares polynomial itself. All these distinctions are recorded in the artifacts.

The final definition is the complete residual list, not merely the existence of an algorithm that could someday print it. The file `direct208/equations/principal208/equation.json` declares each input and each integer coefficient. Its adjacent expansion lists every coefficient of the single quartic. Appendix A prints the residuals.

The simpler 209-variable companion uses the original square-shell syntax and changes only the horizon. Its exact parent-to-child substitution is $t = 2 + 4z$, deleting the unique horizon row and the coordinate t. Three threshold choices, 2, 8, and 16, are retained in `affine209/profiles`; the minimum-2 version has the smallest coefficient height among those three. This companion isolates the horizon improvement from the new sparse-syntax theorem.

5 Weighted square transport: ten positive variables of degree 190

The next equation improves the degree of the inspected ten-positive-variable family. Its exact profile is

$$(\text{variables}, \text{degree}, \text{binary operations}) = (10, 190, 120).$$

The operation split is 75 multiplications, 24 additions, and 21 subtractions; squares and multiplications by constants are charged. This is a tradeoff against the incoming centered free chart's profile $(10, 206, 90)$,

and against source178's scaled whole-product center $(10, 208, 116)$. No global minimum or worldwide priority is asserted. The new step is the polynomial transport weight b^2UE, which permits whole-product centering in the free chart without excluding auxiliary Pell index three.

5.1 The complete polynomial and its fixed data

The independent variables are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The four fixed integers (β, K_0, M_C, M_F) are the complete, unchanged BFS943/radix67 tuple of the frozen source172 parent. The archive supplies all four numerical payloads and reconstructs them from its complete finite 436-character recipe. The largest integer has 592,884 bits. This is the same whole tuple throughout the arithmetic and logical calibration; its scalar bounds alone do not specify the logical instance.

All letters on the following display are polynomial abbreviations, not additional quantified variables:

$$\begin{aligned}
h &= \beta J, & q &= h + 1, & C &= q - F - \alpha,\\
X &= qw, & Y &= s, & \bar{Y} &= qY,\\
k &= \eta + \zeta, & c &= kY + \eta, & A &= \bar{Y}(X + 1) + 1,\\
\Delta &= A^2 - 1, & H &= 4A - 5, & v &= X\bar{Y}^2,\\
\ell &= vk, & E &= 1 + \ell(\ell + k), & U &= ic^2,\\
R &= [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J, & & & &\\
B &= (K_0 + w)C - F, & a &= \Delta U^2 E, & F_0 &= 1 + a,\\
G &= EF_0, & f &= 1 + 2a, & Q &= 4\Delta a F_0,\\
b &= cT - R, & V &= fb - c, & K &= X(3c - 2X) + 2,\\
e_6 &= 6(k - R - 1) - \ell(k^2 - 4), & D &= 6q^7, & M &= hDH.
\end{aligned}$$

Define the following division-free center:

$$\begin{aligned}
e_2 &= (k\bar{Y})^2, & S &= (k\bar{Y})\big[\ell + k - (k\bar{Y})(\bar{Y} + 1)\big],\\
\theta &= U^2 b, & R_c &= \theta(3E + e_2) - (b - c),\\
S_c &= 4E^2 + 3S^2 + e_2^2, & \Omega &= U\big[2SR_c + A(4EV - \theta S_c)\big].
\end{aligned}$$

Finally put

$$\begin{aligned}
L &= b^2UE, && (5)\\
Z_0 &= hD\big[H(n - LB) + K\big] - H\big[DB + h(e_6 - 6q^4 s)\big],\\
Z &= Z_0 - M\Omega,\\
P &= Q(Z^2 - 16M^2GV^2) + 16M^2G - Z^2. && (6)
\end{aligned}$$

The single equation is $P = 0$. Every operation displayed is ordinary integer addition, subtraction, or multiplication with fixed exponents. The literal 120-operation graph is supplied in `positive190/equation.json`. There is no quotient, radical, variable exponent, factorial, sequence predicate, or unknown coefficient inside this polynomial.

Consistency theorem. At the identified inherited whole-tuple calibration, ZFC proves

$$(\neg\exists J, F, \alpha, i, T, w, s, n, \eta, \zeta > 0\ P = 0) \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

Here consistency is finite syntactic non-refutability for the fixed effective ZFC presentation of the parent. The result establishes an equivalence; it does not establish either consistency or unsolvability separately.

5.2 The inherited interface and its precise point of use

The pinned source172 ordinary proof supplies the following named ingredients: prime-valuation square extraction; adjacent-coefficient rational Pell descent; signed all-index folding and normalized strong-index recovery; the full Lucas expansion; the λ-uniform ratio and complete-product projection theorem; and fresh native reconstruction followed by the whole-tuple syntactic refutation calibration. The selected source text is retained as `inherited/FREE_CHART_PROOF.md`, with its original source pins. The new proof does not claim a freshly compiled transitive Lean/Rocq closure.

For proof notation only, let

$$t = \frac{B}{h}, \quad \kappa = \frac{K}{H}, \quad \sigma = \frac{Y}{q^3}, \quad e^* = \frac{e_6}{6}, \quad \rho = \frac{e^*}{q^7}, \quad z = \frac{Z}{M}, \quad W = 1 + hL.$$

These are rational numbers until the appropriate divisibilities are proved. Direct algebra gives

$$z = n + \kappa + \sigma - Wt - \rho - \Omega. \tag{7}$$

The elementary positive-domain bounds are

$$q \geq 32, \quad X \geq q, \quad \bar{Y} \geq q, \quad A > q^2, \quad c \geq 3, \quad U \geq c^2, \quad E > 1, \quad a > c, \quad f > 2c, \quad Q > c^2, \quad M > 0.$$

At a zero, use the exact identities

$$Q = 4\Delta^2 U^2 G, \qquad \gcd(E, F_0) = 1, \qquad (Q-1)Z^2 = 16M^2 G(QV^2 - 1).$$

The prime valuations show that $G = g^2$, $E = \tau^2$, and $F_0 = f_0^2$ for positive integers g, τ, f_0; the added factor16 is a square. The retained adjacent-coefficient descent applies to the initially rational $y = |z|/(4g)$ and proves it is integral. Thus

$$z = \varepsilon 4gy, \quad \varepsilon \in \{-1, 1\}, \qquad QV^2 - (Q-1)y^2 = 1. \tag{8}$$

These arguments depend only on the final norm equation, not the carrier. Signed folding and the first index bound then give, before any assumption on the signs of B, C, R, b, V,

$$\begin{gathered} c = \psi_A(p), \quad p \geq 3 \text{ odd}, \qquad c \geq 4\Delta + 3 > 4q^4, \quad K > 0, \\ b \neq 0, \quad \operatorname{sgn} V = \operatorname{sgn} b, \quad y < \tfrac{3}{2}|V|, \quad |V| < \tfrac{3}{2} f|b|, \quad |b| > 4\Delta f - 1. \end{gathered} \tag{9}$$

These bounds explicitly allow auxiliary index three. Since the final quantities are integers, $|b| \geq 4\Delta f$. Also

$$G = E + \Delta(UE)^2 \leq A^2(UE)^2, \qquad g \leq AUE.$$

5.3 The center bound in the changed chart

The whole-product center of source178 was developed in a different chart. Its polynomial identity is valid here with its letter Y replaced by $\bar{Y} = qY$. The bound must be checked before recovering the sign of B. Temporarily let u be a formal indeterminate, and write

$$\begin{aligned} E(u) &= e_2u^2 + e_1u + e_0, \\ e_1 &= k^2\bar{Y}(1 - 2\bar{Y} - 2\bar{Y}^2), \\ e_0 &= 1 + k^2\bar{Y}(\bar{Y}+1)[\bar{Y}(\bar{Y}+1) - 1], \\ F(u) &= E(u)\big[(1 + 2(u^2-1)U^2E(u))b - c\big]. \end{aligned}$$

For $j = 1, 3$, let Q_j denote the ordinary polynomial quotient of $F(u)$ by u^j, evaluated at $u = A$. This is formal polynomial division, never a floor or rational instruction in the equation. Since

$$F(u) = (b-c)E(u) + 2U^2b(u^2-1)E(u)^2,$$

Q_3 is coefficientwise even. Exact polynomial algebra gives

$$\Omega = U(4AEV - 2Q_1 - Q_3/2). \tag{10}$$

The displayed division-free evaluator is precisely this polynomial. For example, S is the quotient of E by A. If

$$J_1 = S(2E - AS), \quad J_3 = e_2(2S - Ae_2), \quad H_1 = AE^2 - J_1, \quad H_3 = J_1 - J_3,$$

then $Q_1 = (b-c)S + 2\theta H_1$, $Q_3 = 2\theta H_3$, and

$$4H_1 + H_3 = A(4E^2 + 3S^2 + e_2^2) - 2S(3E + e_2).$$

Substitution proves (10) in the free integer polynomial ring.

For a polynomial in u, use a superscript $\#$ for evaluation at A after taking absolute values of all coefficients. Put $\delta = (\bar{Y}+1)/A < 1/16$. The inequalities

$$|e_1| < 2e_2(\bar{Y}+1), \qquad 0 < e_0 \le e_2(\bar{Y}+1)^2$$

give

$$E^{\#}(A) \le e_2(A + \bar{Y} + 1)^2, \quad E \ge e_2A^2(1-2\delta), \quad E^{\#}(A) < 2E.$$

Consequently

$$F^{\#}(A) < 2E(|b| + c) + 8(A^2+1)U^2E^2|b|.$$

At a zero $|b| \ge 1$, $c + 1 < 2U^2E$, and $|V| > \Delta U^2E|b|$, because $f - c > a = \Delta U^2E$. It follows that

$$\frac{F^{\#}(A)}{|EV|} < \frac{4}{\Delta} + \frac{8(A^2+1)}{\Delta} = 8 + \frac{20}{\Delta} < 10.$$

Deletion of low coefficients gives $|Q_j| \le F^{\#}(A)/A^j$. Thus, using $A \ge 16$ in (10),

$$|\Omega - 4AUEV| < 10U|EV|\left(\frac{2}{A} + \frac{1}{2A^3}\right) < AU|EV|.$$

In particular Ω has the sign of V, and the required uniform signed estimate is

$$3AUE|V| < |\Omega| < 5AUE|V|. \tag{11}$$

5.4 The new weighted transport argument

The source178 weight b^2 does not have a proved adequate margin in this free chart when auxiliary index three is possible. The additional factor UE in (5) resolves that problem. Indeed, from (9) and (11),

$$\begin{aligned} 4gy + |\Omega| < 11AUE|V| &< \frac{33}{2}AUEf|b| \\ &\le \frac{33A}{8\Delta}\, b^2UE < \frac{L}{3}. \end{aligned} \tag{12}$$

The last inequality follows from $99A < 8(A^2-1)$ for $A \ge 16$; here in fact $A > 1024$. Thus the proof neither assumes nor needs the stronger scaled-chart index exclusion.

Since $k \ge 2$, the inverse-Lucas correction satisfies

$$e^* \le k - R - 1 < b, \qquad \rho < \frac{|b|}{q^7} < \frac{L}{2}.$$

This holds for every integer R, including zero and negative values. If $B \le -1$, then $Wt = B/h + LB < -L$. Solving (7) for the positive input n and using positive κ, σ gives

$$n = \varepsilon 4gy + \Omega - \kappa - \sigma + Wt + \rho < \frac{L}{3} - L + \frac{L}{2} < 0,$$

a contradiction. Hence $B \ge 0$. Since $F > 0$ and $K_0 + w > 0$, the identity $B = (K_0 + w)C - F$ forces $C > 0$.

Only now invoke the authenticated whole-tuple mask estimate $3q < R < q^4 - 2$. Since $c > 4q^4$, it implies $b, V > 0$. The unchanged normalized strong-index and signed auxiliary-index results identify $p = R$, from

$$V = -c \pmod f, \qquad V = -R \pmod c, \qquad c^2 \mid 2Ug.$$

The retained parity argument yields odd R, even q, and odd J.

5.5 Authentication, projection, and refutation

The only changes in the rational carrier are the integer polynomial Ω and the integer weight L, with $W \equiv 1 \pmod h$. The parent authentication and projection proof therefore applies in the following order. The first Pell norm gives $k = 2\psi_{1+2v}(m)$, $m \ge 2$, and monotonicity gives $m < R$. Integral z and the literal denominator give $q^4 \mid e^*$. The full retained Lucas identity is

$$3\big(e^* - (2m - R - 1)\big) = 4v^2(24Z_m - A_m S_m),$$

where the right side is an integer expression from the finite binomial expansion of the Pell sequence. Since $v = q^3 ws^2$, at every prime dividing q, including3, this implies $q^4 \mid (2m - R - 1)$. Its absolute value is below $R < q^4$, so $m = (R+1)/2$.

Reduction of the literal numerator modulo H gives $H \mid 6hK$. The $\lambda = q$ uniform ratio theorem and the complete-product strict bound force $X = 2^R$ and positive integral κ; this step does not cancel an unproved factor of $6h$. Because $q \mid X$, the radix q is a power of2. Rank growth gives $q^6 \mid X$, hence $q^8 \mid v$. Only now does the same Lucas identity imply $q^7 \mid e^*$, so ρ is integral.

Finally (7), with integral Ω and $W \equiv 1 \pmod h$, gives $q^3 \mid Y$ and $h \mid B$. Thus $t = B/h$ is a nonnegative integer. These are exactly the parent native outer conditions. The inherited fixed-minus parity, half-binomial rounding, and fresh native reconstruction supply the native words at the complete fixed tuple; its named syntactic calibration yields a finite ZFC refutation. This does not replace that calibration by the elementary coefficient inequalities.

5.6 Constructive positive completeness

Conversely, a finite syntactic ZFC refutation supplies the parent canonical outer data and positive η, ζ, i, T. In particular $b, V > 0$, $t \geq 0$ is integral, and ρ, κ, σ are integral. The retained canonical estimates give $g > \kappa + \sigma + |\rho|$. Keep these inputs and define

$$n = 4gy + \Omega + (1 + hL)t + \rho - \kappa - \sigma.$$

Every term is integral; $\Omega > 0$, $y \geq 1$, and $t \geq 0$ make $n > 0$. Its carrier satisfies $z = 4gy$, so (8) makes the literal polynomial vanish. This proves both directions of positive solvability versus finite refutability. The arithmetic arguments and the retained proof-code equivalence are internalizable in ZFC; negation gives the consistency theorem. No restriction to externally standard proof codes is introduced.

5.7 Exact degree and finite verification

The free-chart degree weights for $(A, \bar{Y}, k, U, b, c)$ are $(4, 2, 1, 5, 4, 2)$. Exact sparse collection gives 84 center monomials, degree59, and 594 monomials in $D_{\mathrm{norm}} = \Omega^2 - 16GV^2$, of weighted degree 94. Its entire highest component is

$$[D_{\mathrm{norm}}]_{94} = 8A^6\bar{Y}^8k^8U^6b^2(A^2 - 6\bar{Y}^4).$$

The exact output decomposition is

$$P = Q[Z_0^2 - 2MZ_0\Omega + M^2D_{\mathrm{norm}}] + 16M^2G - (Z_0 - M\Omega)^2.$$

Here

$$\deg(Q, M, G, V, E, \Omega, L, Z_0) = (72, 12, 46, 36, 14, 59, 27, 41).$$

The five respective degree bounds in the decomposition are $154, 184, 190, 70, 142$, so the norm-defect term uniquely controls the top degree. With $q_0 = \beta J$ and $k_0 = \eta + \zeta$, the complete actual highest component is

$$[P]_{190} = 9 \cdot 2^{11}q_0^{74}w^{18}s^{52}i^{10}k_0^{32}(q_0 - F)^2(w^2 - 6s^2). \tag{13}$$

This is nonzero for the actual fixed positive β, proving exact degree 190. The full leading form has 198 collected monomials.

The independent standard-library checker verifies the actual literal graph, all-variable/port/gate liveness, the entire center and norm-defect identities, the complete leading form, all four payload hashes, and 60 deliberately corrupted center operators. Both normal and optimized Python runs pass. The separate frozen numeral recipe reconstructs every coefficient byte. These checks authenticate finite artifacts; the unbounded Pell and logical theorems remain ordinary mathematical proofs with the stated inherited dependencies.

Further research. The weighted transport argument admits other integer polynomial centers with the same uniform $O(AUE|V|)$ signed bound. Reducing the 30-gate center evaluator is a concrete cost target. Reducing the degree requires more than the next term of the same A-Laurent expansion: that term removes the $8A^8\bar{Y}^8k^8U^6b^2$ component but retains a nonzero multiple of $A^6\bar{Y}^{12}k^8U^6b^2$, of the same weighted degree 94. Any proposed division by A or w must be replaced by a proved polynomial identity or an explicitly counted additional witness.

6 Verification, interpretation, and further research

6.1 What the executable checks establish

The archive supplies literal polynomial data, pure Python evaluators, producers, and independent finite checkers. The direct checks collect every quartic coefficient, verify all declared coordinates occur, bind the source changes, and replay the exact projection and coordinate certificates. The compact checks reconstruct the four fixed integers, check the entire arithmetic graph, collect the free center and norm defect, and prove the exact degree through a nonzero full leading homogeneous polynomial. Both ordinary and optimized Python modes are used; substantive acceptance conditions do not vanish with assertions. Targeted corruptions exercise the relevant mathematical comparisons.

The main compact center has 84 monomials in its free polynomial ring. Its norm defect has 594, and weighted degree 94. The resulting degree-190 leading homogeneous form has 198 monomials when expanded in the independent highest-degree inputs. These counts are finite algebraic facts, not probabilistic tests. Auxiliary Pell residue tests, when retained from a parent, are examples and are not zeros of a consistency polynomial.

The immediate incoming direct archive is retained byte for byte under `dependencies/Pell_Growth_210_original.zip`. It supplies the original proofs, source equations, and prior certificates. The new component directories include the specific inputs they need. Their source ledgers pin the snapshot and file hashes, and the release manifest authenticates the delivered bytes. Selected proof-assistant sources are evidence for the named calibration boundary; they are not represented as a newly compiled complete dependency closure.

6.2 Why these are improvements, and what they do not settle

The new direct equation reduces the number of unrestricted integer unknowns while retaining degree four. The new compact equation lowers the exact degree at ten positive unknowns. Its extra operations pay for a stronger signed transport bound and an exact center; they are included in the count. Both are concrete tradeoffs against inspected predecessors. Earlier ProveIt files already contain fewer-variable representations in other regimes, so the direct count must not be described as a record over all Diophantine encodings.

The finite-marker theorem is useful beyond this particular count. It separates finite proof transport from a global bijection of all codes. It permits the arithmetic of the constructors to be optimized while unused numbers are handled uniformly as temporary atoms in each finite proof. This is a rigorously justified design freedom, even when one particular sparse code yields only a modest numerical improvement.

6.3 Further research questions

1. **Optimize proof syntax under finite-marker equivalence.** Can a sparse constructor family create more shared quadratic expressions than the four offsets in (1)? An injective partial coding with strict argument descent is enough; surjectivity need not be restored. Any new family must still preserve the designated truth and false codes, the guarded schemata, and the updated growth bounds.
2. **Change the calculus, not just its numeral coding.** The largest term is the guarded seed bundle. A shorter calibrated schema could reduce its depth and the surrounding arithmetic at once. This requires an actual two-way finite-refutation proof; a smaller local expression alone is insufficient.

3. **Couple the transport weight to a sharper center.** For the free chart, can one lower the weighted norm-defect degree below 94 while keeping both the early signed bound and positive completeness? Merely removing the index-three case by assumption would invalidate the present proof. Higher quotient corrections should be compared with the extra degree they introduce through the carrier.
4. **Improve the small-coefficient quartic tradeoff.** The affine slice removes a coordinate but increases coefficient height. Different primitive affine charts or shared divisibility coordinates may retain the reduction with a smaller height. A successful change should report the collected polynomial's height, not only circuit literals.
5. **Reduce verification dependencies.** The new sparse recoding is a short finite-tree argument and is a suitable isolated formalization target if proof-assistant work is resumed. The weighted transport inequality can also be formalized separately from the much larger inherited native calibration.

References

[1] Vladimir Reshetnikov, *ProveIt*, snapshot `d123610bc98fef2006bff1c9466bc3f1221003e7`, https://github.com/VladimirReshetnikov/ProveIt/tree/d123610bc98fef2006bff1c9466bc3f1221003e7.

[2] *Pell growth from divisibility: an explicit quartic for Con(ZFC)*, incoming archive `ConZFC_Pell_Growth_210_2026-10-25.zip` at the snapshot above; the complete archive is retained with this paper. Its standalone growth and affine-radix proofs are the immediate arithmetic dependencies of the direct construction.

[3] *A reused CRT index: an explicit Diophantine consistency equation*, package 173 in `Computability/DiophantineConsistency/con-zfc-diophantine-forms`. This supplies the sharp label-product separation and reused-index anchor.

[4] *Pooled powers, round 21*, package 172, free-chart and Lucas-correction proofs, with the complete pinned native coefficient tuple. The relevant original text and literal parent are retained under `positive190/inherited`.

[5] *Polynomial centers for Diophantine consistency*, package 178, whole-product center and polynomial quotient identity. The relevant original proof and algebra are retained under `positive190/inherited`.

[6] *Signed horizons and residue-encoded Pell witnesses*, incoming archive `ConZFC_Signed_Horizons_and_Residue_Carriers_2026-10-25.zip` at the snapshot above. Signed affine horizons are credited to this predecessor; their composition with the divisibility-growth sign bootstrap is proved here.

A Complete literal residuals of the direct quartic

Every coordinate $x_0, \dots, x_{207}$ is an unrestricted integer. The single equation is

$$\sum_{j=0}^{147} R_j^2 = 0.$$

All coefficients below are integers. This list and the machine-readable residual file define exactly the same polynomial. Subscripts are coordinate indices and fixed exponents are literal exponents.

$R_0 = 2 - 3x_3 - 4x_3x_{108} + x_6 + 4x_{108}$

$$
\begin{aligned}
R_{1} &= -x_{1} - x_{3}x_{4} + x_{5}x_{6} \\
R_{2} &= -1 + x_{0} - x_{5}x_{6} \\
R_{3} &= -2x_{1}x_{135} - x_{2}x_{17} + x_{3} + 2x_{15} \\
R_{4} &= -x_{3} - 2x_{3}x_{15} + 2x_{15} + x_{20} \\
R_{5} &= -x_{13}x_{20} + x_{21} \\
R_{6} &= -2x_{0}^{2} + 2x_{0}x_{11} + 2x_{0}x_{12} + 2x_{0}x_{21} + 2x_{0}x_{22} - x_{11}^{2} - 2x_{11}x_{21} - x_{12}^{2} \\
&\quad - 2x_{12}x_{22} + x_{16}x_{23} - x_{18}x_{19} - x_{21}^{2} - x_{22}^{2} \\
R_{7} &= -x_{3}x_{14} + x_{14} - x_{14}x_{20} + x_{22} \\
R_{8} &= -x_{24}x_{27} + x_{27} + x_{29} \\
R_{9} &= -x_{2}x_{50} + x_{25}x_{123} - x_{28} - x_{29}x_{50} \\
R_{10} &= x_{30}x_{123} - x_{31}x_{53} - x_{32} \\
R_{11} &= x_{33}x_{123} - x_{34}x_{53} - x_{35} \\
R_{12} &= x_{36}x_{123} - x_{37}x_{53} - x_{38} \\
R_{13} &= x_{39}x_{123} - x_{40}x_{53} - x_{41} \\
R_{14} &= x_{42}x_{124} - x_{44}x_{53} - x_{45} \\
R_{15} &= x_{46}x_{124} - x_{47}x_{53} - x_{48} \\
R_{16} &= -x_{16}x_{50} + x_{53} \\
R_{17} &= x_{49} - x_{51}x_{53} - x_{52} \\
R_{18} &= -x_{53}x_{55} + x_{54} - x_{56} \\
R_{19} &= -x_{53}x_{58} + x_{57}x_{123} - x_{59} \\
R_{20} &= x_{60} - 2x_{60}x_{62} + x_{63} \\
R_{21} &= -x_{60}x_{63} - x_{63} + x_{64} \\
R_{22} &= -1 + x_{61}^{2} + x_{63}^{2} - x_{64}^{2} \\
R_{23} &= 1 - x_{61}x_{66} - x_{65} + x_{67} \\
R_{24} &= -1 + 2x_{65} - x_{65}^{2} + 2x_{65}x_{67} + x_{66}^{2} \\
R_{25} &= 1 + x_{2}^{2} - 2x_{2}x_{61} + x_{68} \\
R_{26} &= -x_{2}x_{100} + x_{16} + x_{68}x_{69} - x_{99} \\
R_{27} &= -5 - 8x_{1} - 4x_{1}^{2} - 16x_{1}x_{15} + 4x_{1}x_{61} - 16x_{15} - 16x_{15}^{2} + 8x_{15}x_{61} + 4x_{61} - x_{70} \\
R_{28} &= -1 - 2x_{1}x_{100} - 4x_{15}x_{100} + x_{24} + x_{70}x_{71} - x_{99} - 2x_{100} \\
R_{29} &= -x_{61}x_{73} - x_{72} + x_{74} \\
R_{30} &= -1 - x_{72}^{2} + 2x_{72}x_{74} + x_{73}^{2} \\
R_{31} &= -x_{61}x_{76} - x_{75} + x_{77} \\
R_{32} &= -1 - x_{75}^{2} + 2x_{75}x_{77} + x_{76}^{2} \\
R_{33} &= -x_{61}x_{79} - x_{78} + x_{80}
\end{aligned}
$$

$$
\begin{aligned}
R_{34} &= -1 - x_{78}^2 + 2x_{78}x_{80} + x_{79}^2 \\
R_{35} &= -x_{61}x_{82} - x_{81} + x_{83} \\
R_{36} &= -1 - x_{81}^2 + 2x_{81}x_{83} + x_{82}^2 \\
R_{37} &= -x_{61}x_{85} - x_{84} + x_{86} \\
R_{38} &= -1 - x_{84}^2 + 2x_{84}x_{86} + x_{85}^2 \\
R_{39} &= -x_{61}x_{88} - x_{87} + x_{89} \\
R_{40} &= -1 - x_{87}^2 + 2x_{87}x_{89} + x_{88}^2 \\
R_{41} &= 1 - x_{61}x_{91} - x_{90} + x_{92} \\
R_{42} &= -1 + 2x_{90} - x_{90}^2 + 2x_{90}x_{92} + x_{91}^2 \\
R_{43} &= -x_{61}x_{94} - x_{93} + x_{95} \\
R_{44} &= -1 - x_{93}^2 + 2x_{93}x_{95} + x_{94}^2 \\
R_{45} &= -x_{61}x_{97} - x_{96} + x_{98} \\
R_{46} &= -1 - x_{96}^2 + 2x_{96}x_{98} + x_{97}^2 \\
R_{47} &= 1 - x_{61}x_{100} - x_{99} + x_{101} \\
R_{48} &= -1 + 2x_{99} - x_{99}^2 + 2x_{99}x_{101} + x_{100}^2 \\
R_{49} &= -x_{61}x_{103} - x_{102} + x_{104} \\
R_{50} &= -1 - x_{102}^2 + 2x_{102}x_{104} + x_{103}^2 \\
R_{51} &= -x_{61}x_{106} - x_{105} + x_{107} \\
R_{52} &= -1 - x_{105}^2 + 2x_{105}x_{107} + x_{106}^2 \\
R_{53} &= -28 - x_0^2 - 12x_1 - 6x_1^2 - 16x_1x_{15} - 8x_1x_{108} - x_2^2 - x_3^2 - x_7^2 - x_8^2 - x_9^2 \\
&\quad - x_{10}^2 - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - 16x_{15} - 20x_{15}^2 - x_{16}^2 + 2x_{24} - 2x_{24}^2 \\
&\quad - x_{25}^2 - x_{26}^2 - x_{30}^2 - x_{33}^2 - x_{36}^2 - x_{39}^2 - x_{42}^2 - x_{43}^2 - x_{46}^2 - x_{49}^2 \\
&\quad - x_{50}^2 - x_{54}^2 - x_{57}^2 + x_{60} - 72x_{108} - 64x_{108}^2 \\
R_{54} &= -x_{61}x_{110} + x_{119} \\
R_{55} &= -1 + x_{109}^2 + x_{110}^2 - x_{119}^2 \\
R_{56} &= -x_{112}x_{113} + x_{120} \\
R_{57} &= -1 + x_{111}^2 + x_{112}^2 - x_{120}^2 \\
R_{58} &= -x_{106}^2 + x_{121} \\
R_{59} &= x_{110} - 2x_{115}x_{121} - x_{121} \\
R_{60} &= -1 - 4x_{110} - 8x_{110}x_{114} + x_{113} \\
R_{61} &= -x_{61} - x_{109}x_{116} + x_{113} \\
R_{62} &= -x_{107} - x_{109}x_{117} + x_{111} \\
R_{63} &= -2x_{15} - 4x_{106}x_{118} + x_{112}
\end{aligned}
$$

$$
\begin{aligned}
R_{64} = {} & 17 + x_0^2 - 2x_0x_{100} + 4x_1 + x_1^2 - 2x_1x_{73} + 8x_1x_{108} + x_3^2 - 2x_3x_{91} + x_7^2 \\
& - 2x_7x_{76} + x_8^2 - 2x_8x_{79} + x_9^2 - 2x_9x_{82} + x_{10}^2 - 2x_{10}x_{85} + x_{11}^2 - 2x_{11}x_{88} \\
& + x_{12}^2 - 2x_{12}x_{94} + x_{13}^2 - 2x_{13}x_{97} + x_{14}^2 - 2x_{14}x_{103} - 4x_{15}x_{106} - x_{61}x_{122} \\
& - 4x_{66} - 8x_{66}x_{108} + x_{67}^2 - 4x_{73} - 8x_{73}x_{108} - 2x_{74} + x_{74}^2 - 2x_{77} + x_{77}^2 - 2x_{80} \\
& + x_{80}^2 - 2x_{83} + x_{83}^2 - 2x_{86} + x_{86}^2 - 2x_{89} + x_{89}^2 + x_{92}^2 - 2x_{95} + x_{95}^2 - 2x_{98} \\
& + x_{98}^2 + x_{101}^2 - 2x_{104} + x_{104}^2 + 2x_{106}^2 - 2x_{107} + x_{107}^2 + 32x_{108} + 32x_{108}^2 + x_{122} \\
& - x_{125}^2 - x_{126}^2 - x_{127} - x_{127}^2 \\
R_{65} = {} & - x_{26}x_{123} + x_{50} \\
R_{66} = {} & - x_{43}x_{124} + x_{50} \\
R_{67} = {} & - 1 - 4x_{15}^2 - x_{66}^2 - x_{73}^2 - x_{76}^2 - x_{79}^2 - x_{82}^2 - x_{85}^2 - x_{88}^2 - x_{91}^2 \\
& - x_{94}^2 - x_{97}^2 - x_{100}^2 - x_{103}^2 + x_{106}^2 - x_{125}^2 - x_{126}^2 - x_{127} - x_{127}^2 \\
R_{68} = {} & - 3 - 4x_{28}^2 - 4x_{32}^2 - 4x_{35}^2 - 4x_{38}^2 - 4x_{41}^2 - 4x_{45}^2 - 4x_{48}^2 + x_{50}^2 - 4x_{52}^2 \\
& - 4x_{56}^2 - 4x_{59}^2 - x_{128}^2 - x_{129}^2 + 3x_{130} - x_{130}^2 \\
R_{69} = {} & x_{24}x_{132} - 2x_{25}x_{73} - 2x_{30}x_{76} - 2x_{33}x_{79} - 2x_{36}x_{82} - 2x_{39}x_{85} - 2x_{42}x_{88} \\
& - 2x_{46}x_{94} - 2x_{49}x_{97} - 2x_{54}x_{103} - 2x_{57}x_{106} - 2x_{61}x_{132} + 2x_{73}x_{74} + 2x_{76}x_{77} \\
& + 2x_{79}x_{80} + 2x_{82}x_{83} + 2x_{85}x_{86} + 2x_{88}x_{89} + 2x_{94}x_{95} + 2x_{97}x_{98} + 2x_{103}x_{104} \\
& + 2x_{106}x_{107} + x_{131} \\
R_{70} = {} & - 10 - x_{24}x_{131} + x_{25}^2 - 2x_{25}x_{72} + x_{30}^2 - 2x_{30}x_{75} + x_{33}^2 - 2x_{33}x_{78} + x_{36}^2 \\
& - 2x_{36}x_{81} + x_{39}^2 - 2x_{39}x_{84} + x_{42}^2 - 2x_{42}x_{87} + x_{46}^2 - 2x_{46}x_{93} + x_{49}^2 \\
& - 2x_{49}x_{96} + x_{54}^2 - 2x_{54}x_{102} + x_{57}^2 - 2x_{57}x_{105} + 2x_{72}x_{74} + 2x_{75}x_{77} + 2x_{78}x_{80} \\
& + 2x_{81}x_{83} + 2x_{84}x_{86} + 2x_{87}x_{89} + 2x_{93}x_{95} + 2x_{96}x_{98} + 2x_{102}x_{104} + 2x_{105}x_{107} + x_{132} \\
R_{71} = {} & x_{24}x_{134} - 2x_{26}x_{66} - 2x_{43}x_{91} - 2x_{50}x_{100} - 2x_{61}x_{134} + 2x_{66} + 2x_{66}x_{67} + 2x_{91} \\
& + 2x_{91}x_{92} + 2x_{100} + 2x_{100}x_{101} + x_{133} - x_{134} \\
R_{72} = {} & - 2 - x_{24}x_{133} + x_{26}^2 - 2x_{26}x_{65} + x_{43}^2 - 2x_{43}x_{90} + x_{50}^2 - 2x_{50}x_{99} + 2x_{65} \\
& + 2x_{65}x_{67} + 2x_{90} + 2x_{90}x_{92} + x_{99}^2 - x_{100}^2 + x_{133} + x_{134} \\
R_{73} = {} & - 1 - x_1 - 2x_1x_{135} + x_3 \\
R_{74} = {} & - 2x_1 - 4x_1x_{108} + x_{137} \\
R_{75} = {} & - 1 - 16x_1x_{137} + x_{136}^2 - 64x_{137}^2 \\
R_{76} = {} & - 1048576 + x_1 - 2097152x_{108} - 4194304x_{108}x_{138} - 2097152x_{138} \\
R_{77} = {} & - x_7^2 - 2x_7x_8 - x_8^2 + x_{139} \\
R_{78} = {} & - 16 - 32x_7 - 16x_7^2 - 32x_7x_{139} - 32x_{139} - 16x_{139}^2 + x_{140} \\
R_{79} = {} & - 144 - 408x_7 - 289x_7^2 - 544x_7x_{139} - 136x_7x_{140} - 384x_{139} - 256x_{139}^2 - 128x_{139}x_{140} \\
& - 96x_{140} - 16x_{140}^2 + x_{141} \\
R_{80} = {} & - 9 - 24x_7 - 16x_7^2 - 32x_7x_{141} - 24x_{141} - 16x_{141}^2 + x_{142} \\
R_{81} = {} & - x_8^2 - 2x_8x_9 - x_9^2 + x_{143} \\
R_{82} = {} & - x_7^2 - 2x_7x_9 - x_9^2 + x_{144}
\end{aligned}
$$

$$
\begin{aligned}
R_{83} = & -16-64x_{7}-65x_{7}^{2}-8x_{7}x_{8}-64x_{7}x_{139}-8x_{7}x_{143}-64x_{7}x_{144}-16x_{8}^{2}-32x_{8}x_{143}\\
& -32x_{139}-16x_{139}^{2}-32x_{139}x_{144}-16x_{143}^{2}-32x_{144}-16x_{144}^{2}+x_{145}\\
R_{84} = & -100-400x_{7}-404x_{7}^{2}-32x_{7}x_{8}-640x_{7}x_{139}-32x_{7}x_{143}-160x_{7}x_{145}-64x_{8}^{2}\\
& -128x_{8}x_{143}-320x_{139}-256x_{139}^{2}-128x_{139}x_{145}-64x_{143}^{2}-80x_{145}-16x_{145}^{2}+x_{146}\\
R_{85} = & -4-32x_{9}-64x_{9}^{2}-64x_{9}x_{143}-64x_{9}x_{144}-16x_{143}-16x_{143}^{2}-32x_{143}x_{144}-16x_{144}\\
& -16x_{144}^{2}+x_{147}\\
R_{86} = & -49-56x_{7}-16x_{7}^{2}-128x_{7}x_{9}-32x_{7}x_{139}-128x_{7}x_{144}-32x_{7}x_{147}-224x_{9}-256x_{9}^{2}\\
& -128x_{9}x_{139}-512x_{9}x_{144}-128x_{9}x_{147}-56x_{139}-16x_{139}^{2}-128x_{139}x_{144}-32x_{139}x_{147}\\
& -224x_{144}-256x_{144}^{2}-128x_{144}x_{147}-56x_{147}-16x_{147}^{2}+x_{148}\\
R_{87} = & -4-16x_{7}-17x_{7}^{2}-16x_{7}x_{8}-8x_{7}x_{143}-32x_{7}x_{144}-4x_{8}-17x_{8}^{2}-32x_{8}x_{143}\\
& -8x_{8}x_{144}-16x_{143}^{2}-16x_{144}-16x_{144}^{2}+x_{149}\\
R_{88} = & -400-800x_{7}-404x_{7}^{2}-2752x_{7}x_{8}-2592x_{7}x_{143}-160x_{7}x_{149}-2720x_{8}-4688x_{8}^{2}\\
& -8832x_{8}x_{143}-544x_{8}x_{149}-2560x_{143}-4160x_{143}^{2}-512x_{143}x_{149}-160x_{149}-16x_{149}^{2}\\
& +x_{150}\\
R_{89} = & -4-16x_{7}-16x_{7}^{2}-32x_{7}x_{9}-32x_{7}x_{139}-32x_{7}x_{143}-16x_{9}-16x_{9}^{2}-32x_{9}x_{139}\\
& -32x_{9}x_{143}-16x_{139}-16x_{139}^{2}-32x_{139}x_{143}-16x_{143}-16x_{143}^{2}+x_{151}\\
R_{90} = & -49-280x_{9}-400x_{9}^{2}-640x_{9}x_{143}-160x_{9}x_{144}-160x_{9}x_{151}-224x_{143}-256x_{143}^{2}\\
& -128x_{143}x_{144}-128x_{143}x_{151}-56x_{144}-16x_{144}^{2}-32x_{144}x_{151}-56x_{151}-16x_{151}^{2}+x_{152}\\
R_{91} = & -400-600x_{7}-225x_{7}^{2}-480x_{7}x_{139}+120x_{7}x_{140}-640x_{139}-256x_{139}^{2}+128x_{139}x_{140}\\
& +160x_{140}-16x_{140}^{2}+x_{153}\\
R_{92} = & -9-24x_{7}-16x_{7}^{2}-32x_{7}x_{153}-24x_{153}-16x_{153}^{2}+x_{154}\\
R_{93} = & -64-256x_{7}-256x_{7}^{2}-32x_{7}x_{8}-512x_{7}x_{153}-128x_{7}x_{154}-16x_{8}-x_{8}^{2}-32x_{8}x_{153}\\
& -8x_{8}x_{154}-256x_{153}-256x_{153}^{2}-128x_{153}x_{154}-64x_{154}-16x_{154}^{2}+x_{155}\\
R_{94} = & -33489-87840x_{7}-57600x_{7}^{2}+1920x_{7}x_{9}-115200x_{7}x_{139}+7680x_{7}x_{140}+1920x_{7}x_{143}+1464x_{9}\\
& -16x_{9}^{2}+1920x_{9}x_{139}-128x_{9}x_{140}-32x_{9}x_{143}-87840x_{139}-57600x_{139}^{2}+7680x_{139}x_{140}\\
& +1920x_{139}x_{143}+5856x_{140}-256x_{140}^{2}-128x_{140}x_{143}+1464x_{143}-16x_{143}^{2}+x_{156}\\
R_{95} = & -225-600x_{9}-400x_{9}^{2}-640x_{9}x_{143}-160x_{9}x_{144}-160x_{9}x_{156}-480x_{143}-256x_{143}^{2}\\
& -128x_{143}x_{144}-128x_{143}x_{156}-120x_{144}-16x_{144}^{2}-32x_{144}x_{156}-120x_{156}-16x_{156}^{2}\\
& +x_{157}\\
R_{96} = & -x_{9}^{2}-2x_{9}x_{10}-x_{10}^{2}+x_{158}\\
R_{97} = & -17x_{7}^{2}-32x_{7}x_{8}-8x_{7}x_{9}-32x_{7}x_{143}-32x_{7}x_{144}-8x_{7}x_{158}-16x_{8}^{2}-32x_{8}x_{143}\\
& -32x_{8}x_{144}-16x_{9}^{2}-32x_{9}x_{158}-16x_{143}^{2}-32x_{143}x_{144}-16x_{144}^{2}-16x_{158}^{2}+x_{159}\\
R_{98} = & -260x_{7}^{2}-544x_{7}x_{9}-512x_{7}x_{144}-544x_{7}x_{158}-128x_{7}x_{159}-320x_{9}^{2}-512x_{9}x_{144}\\
& -640x_{9}x_{158}-128x_{9}x_{159}-256x_{144}^{2}-512x_{144}x_{158}-128x_{144}x_{159}-320x_{158}^{2}\\
& -128x_{158}x_{159}-16x_{159}^{2}+x_{160}
\end{aligned}
$$

$$
\begin{aligned}
R_{99} = &-16-136x_{9}-289x_{9}^{2}-544x_{9}x_{144}-136x_{9}x_{147}-128x_{144}-256x_{144}^{2}-128x_{144}x_{147}\\
&-32x_{147}-16x_{147}^{2}+x_{161}\\
R_{100} = &-4-16x_{8}-16x_{8}^{2}-32x_{8}x_{143}-16x_{143}-16x_{143}^{2}+x_{162}\\
R_{101} = &-16-128x_{8}-256x_{8}^{2}-32x_{8}x_{9}-512x_{8}x_{143}-128x_{8}x_{162}-8x_{9}-x_{9}^{2}-32x_{9}x_{143}\\
&-8x_{9}x_{162}-128x_{143}-256x_{143}^{2}-128x_{143}x_{162}-32x_{162}-16x_{162}^{2}+x_{163}\\
R_{102} = &-x_{8}^{2}-8x_{8}x_{9}-8x_{8}x_{163}-16x_{9}^{2}-32x_{9}x_{163}-16x_{163}^{2}+x_{164}\\
R_{103} = &-9-24x_{8}-16x_{8}^{2}-32x_{8}x_{164}-24x_{164}-16x_{164}^{2}+x_{165}\\
R_{104} = &-64-256x_{8}-256x_{8}^{2}-32x_{8}x_{9}-512x_{8}x_{164}-128x_{8}x_{165}-16x_{9}-x_{9}^{2}-32x_{9}x_{164}\\
&-8x_{9}x_{165}-256x_{164}-256x_{164}^{2}-128x_{164}x_{165}-64x_{165}-16x_{165}^{2}+x_{166}\\
R_{105} = &-x_{8}^{2}-8x_{8}x_{9}-8x_{8}x_{166}-16x_{9}^{2}-32x_{9}x_{166}-16x_{166}^{2}+x_{167}\\
R_{106} = &-16-32x_{8}-16x_{8}^{2}-32x_{8}x_{9}-32x_{8}x_{161}-32x_{8}x_{167}-32x_{9}-16x_{9}^{2}-32x_{9}x_{161}\\
&-32x_{9}x_{167}-32x_{161}-16x_{161}^{2}-32x_{161}x_{167}-32x_{167}-16x_{167}^{2}+x_{168}\\
R_{107} = &-64-16x_{7}-x_{7}^{2}-32x_{7}x_{9}-32x_{7}x_{161}-8x_{7}x_{168}-256x_{9}-256x_{9}^{2}-512x_{9}x_{161}\\
&-128x_{9}x_{168}-256x_{161}-256x_{161}^{2}-128x_{161}x_{168}-64x_{168}-16x_{168}^{2}+x_{169}\\
R_{108} = &-9-24x_{7}-16x_{7}^{2}-32x_{7}x_{169}-24x_{169}-16x_{169}^{2}+x_{170}\\
R_{109} = &-64-256x_{7}-256x_{7}^{2}-32x_{7}x_{9}-512x_{7}x_{169}-128x_{7}x_{170}-16x_{9}-x_{9}^{2}-32x_{9}x_{169}\\
&-8x_{9}x_{170}-256x_{169}-256x_{169}^{2}-128x_{169}x_{170}-64x_{170}-16x_{170}^{2}+x_{171}\\
R_{110} = &-x_{7}^{2}-8x_{7}x_{9}-8x_{7}x_{171}-16x_{9}^{2}-32x_{9}x_{171}-16x_{171}^{2}+x_{172}\\
R_{111} = &-9-24x_{7}-16x_{7}^{2}-32x_{7}x_{172}-24x_{172}-16x_{172}^{2}+x_{173}\\
R_{112} = &-81-288x_{7}-256x_{7}^{2}-128x_{7}x_{8}-128x_{7}x_{143}-512x_{7}x_{172}-128x_{7}x_{173}-72x_{8}-16x_{8}^{2}\\
&-32x_{8}x_{143}-128x_{8}x_{172}-32x_{8}x_{173}-72x_{143}-16x_{143}^{2}-128x_{143}x_{172}-32x_{143}x_{173}\\
&-288x_{172}-256x_{172}^{2}-128x_{172}x_{173}-72x_{173}-16x_{173}^{2}+x_{174}\\
R_{113} = &-25-160x_{8}-305x_{8}^{2}-544x_{8}x_{143}-32x_{8}x_{167}-136x_{8}x_{174}-128x_{143}-256x_{143}^{2}\\
&-128x_{143}x_{174}-24x_{167}-16x_{167}^{2}-32x_{174}-16x_{174}^{2}+x_{175}\\
R_{114} = &-64-304x_{8}-336x_{8}^{2}-512x_{8}x_{167}-160x_{8}x_{175}-224x_{167}-192x_{167}^{2}-128x_{167}x_{175}\\
&-84x_{175}-16x_{175}^{2}+x_{176}\\
R_{115} = &-73-24x_{7}-16x_{7}^{2}-32x_{7}x_{9}-32x_{7}x_{139}-32x_{7}x_{178}-256x_{8}-256x_{8}^{2}-32x_{8}x_{9}\\
&-128x_{8}x_{176}-40x_{9}-17x_{9}^{2}-32x_{9}x_{139}-8x_{9}x_{176}-32x_{9}x_{178}-24x_{139}-16x_{139}^{2}\\
&-32x_{139}x_{178}-64x_{176}-16x_{176}^{2}+x_{177}-24x_{178}-16x_{178}^{2}\\
R_{116} = &-1-8x_{7}-16x_{7}^{2}-8x_{7}x_{9}-32x_{7}x_{139}-2x_{9}-x_{9}^{2}-8x_{9}x_{139}-8x_{139}-16x_{139}^{2}+x_{178}\\
R_{117} = &-72-288x_{7}-320x_{7}^{2}-256x_{7}x_{9}-640x_{7}x_{139}-128x_{7}x_{177}-128x_{7}x_{178}-144x_{9}-80x_{9}^{2}\\
&-256x_{9}x_{139}-32x_{9}x_{177}-128x_{9}x_{178}-288x_{139}-320x_{139}^{2}-128x_{139}x_{177}-128x_{139}x_{178}\\
&-48x_{177}-16x_{177}^{2}-96x_{178}-64x_{178}^{2}+x_{179}\\
R_{118} = &-289-2176x_{7}-4096x_{7}^{2}-8192x_{7}x_{139}-512x_{7}x_{179}-2176x_{139}-4096x_{139}^{2}-512x_{139}x_{179}\\
&-136x_{179}-16x_{179}^{2}+x_{180}
\end{aligned}
$$

$$R_{119} = -x_{1}x_{12} + x_{181}$$

$$R_{120} = -x_{1}x_{11} + x_{182}$$

$$R_{121} = 16x_{7}x_{181} - x_{11} + x_{12} + 16x_{141}x_{181} + 4x_{142}x_{181} + 8x_{181} + x_{183}$$

$$R_{122} = 16x_{7}x_{181} - x_{11} + x_{12} + 4x_{146}x_{181} + 8x_{181} + x_{184}$$

$$R_{123} = 16x_{7}x_{181} - x_{11} + x_{12} + 16x_{139}x_{181} + 4x_{148}x_{181} + 12x_{181} + x_{185}$$

$$R_{124} = 80x_{7}x_{181} + 256x_{8}x_{181} - x_{11} + x_{12} + 256x_{143}x_{181} + 4x_{150}x_{181} + 72x_{181} + x_{186}$$

$$R_{125} = 16x_{9}x_{181} - x_{11} + x_{12} + 16x_{144}x_{181} + 4x_{152}x_{181} + 12x_{181} + x_{187}$$

$$R_{126} = 64x_{7}x_{181} - x_{11} + x_{12} + 64x_{153}x_{181} + 16x_{154}x_{181} + 4x_{155}x_{181} + 32x_{181} + x_{188}$$

$$R_{127} = 16x_{9}x_{181} - x_{11} + x_{12} + 16x_{144}x_{181} + 4x_{157}x_{181} + 12x_{181} + x_{189}$$

$$R_{128} = 64x_{9}x_{181} - x_{11} + x_{12} + 64x_{158}x_{181} + 4x_{160}x_{181} + x_{190}$$

$$R_{129} = -x_{7}x_{8} + x_{7}x_{9} + x_{8}x_{9} - x_{9}^{2} + x_{191}$$

$$R_{130} = 1 + 2x_{10} + x_{10}^{2} - x_{191}^{2} + x_{192}$$

$$R_{131} = 256x_{7}x_{181} - x_{11} + x_{12} + 256x_{139}x_{181} + 16x_{179}x_{181} + 4x_{180}x_{181} + 64x_{181} + x_{193}$$

$$R_{132} = x_{18} - x_{192}^{2} - x_{193}^{2}$$

$$R_{133} = -1 - 4x_{1}x_{7} - 4x_{1}x_{139} + x_{194}$$

$$R_{134} = -x_{7}x_{182} - x_{11} + x_{195}$$

$$R_{135} = -x_{8}x_{181} - x_{12} + x_{194}x_{195} + x_{196}$$

$$R_{136} = 4x_{7}x_{181} - x_{8}x_{182} - x_{11} + x_{12} + 4x_{139}x_{181} + 2x_{181} + x_{197}$$

$$R_{137} = -x_{11}x_{183} + x_{12}x_{183} + x_{198}$$

$$R_{138} = -x_{184}x_{198} + x_{199}$$

$$R_{139} = -x_{185}x_{199} + x_{200}$$

$$R_{140} = -x_{186}x_{200} + x_{201}$$

$$R_{141} = -x_{187}x_{201} + x_{202}$$

$$R_{142} = -x_{188}x_{202} + x_{203}$$

$$R_{143} = -x_{189}x_{203} + x_{204}$$

$$R_{144} = -x_{190}x_{204} + x_{205}$$

$$R_{145} = -x_{196}x_{205} + x_{206}$$

$$R_{146} = -x_{197}x_{206} + x_{207}$$

$$R_{147} = x_{19} - x_{207}^{2}$$

B Complete fixed coefficients for the ten-variable polynomial

B.1 The complete fixed coefficient recipe

All quantities in this subsection are part of a finite specification of fixed integers; none is an additional unknown in the polynomial. The symbol K in this inherited recipe denotes our fixed port K_0; the temporary B below is local to this recipe. The calibrated tuple is copied as a whole.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{i=0}^{942} v(t_i) x^i \\ &\quad + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} &= \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ &\quad (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ &\quad (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\ \mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\ r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w, n, e, s)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j+2\ (j = 2q + b < 430)$	$((1-d_j)l_{u_j}, h(q,b), d_j r_{u_j}, v_j)$
$432 + b\ (b = 0, 1)$	$(0, h(215, b), 0, h(215, b))$
$434 + b\ (b = 0, 1)$	$(0, b, 0, b)$
$424 + 2r_q + b\ (q \in \mathcal{R}, b = 0, 1)$	$(r_q, b, 0, h(q, b))$
$424 + 2l_q + b\ (q \in \mathcal{L}, b = 0, 1)$	$(0, b, l_q, h(q, b))$
$924 + j\ (0 \le j < 4)$	$(0, j + 430\lfloor j/2 \rfloor, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$^{934+3a+b-6\lfloor a/4 \rfloor}_{a\in\{0,1,4\},\ b=0,1,2}$	$(a, 437, b, 437)$

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSD0Tgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

C Reproduction and artifact map

From the extracted package root, run

```
python3 -B verify_all.py
python3 -O -B verify_all.py
```

The programs require Python 3.10 or later and its standard library. They do not search for roots. An evaluator accepts an explicitly provided integer tuple in the order declared by its equation file. Exact evaluation may produce large integers; a modular diagnostic is not a proof of an integer zero. The paper can be rebuilt with a normal TeX Live installation:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

Mathematical data are deterministic. PDF bytes can vary with TeX version, font metadata, or timestamps. The release manifest records the actual delivered files and should be regenerated after an intentional edit.

Path	Contents
`report/`	This paper, its editable LaTeX, printed residuals, and metrics.
`direct208/`	New literal quartic, expansion, constructor producer, projection and optimization certificates, exact source audit, and evaluator.
`positive190/`	Ten-variable graph, whole coefficient payloads and reconstruction, complete new proof, selected parent proofs, and independent finite checker.
`affine209/`	Three isolated affine-horizon profiles, parent coefficients, source commutation certificates, and independent checks.
`reviews/`	Separately assigned internal mathematical and algebraic reviews; these are AI-assisted reviews, not external human peer review.
`dependencies/`	Unchanged incoming Pell Growth 210 archive.
`verify_all.py`	Ordered, offline verification of the delivered components.